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BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : LunaForge

Team Leader Name : Aditya Narayan

Problem Statement : Challenge 8 – Detection and Characterization of Subsurface Ice in Lunar South Polar Regions Using Chandrayaan-2 Radar and Imagery Data for Landing Site and Rover Traverse Planning

Team Members

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LIRAF: Problem, Solution & Innovations

1. Problem

- Lunar south pole contains PSRs with potential water ice.
- Existing datasets provide complementary but incomplete information.
- Current approaches analyse datasets independently.

2. Our Solution

LIRAF integrates radar, terrain, hydrogen abundance, illumination, and OHRC imagery into a unified decision-support framework for ice detection and mission planning.

3. Key Innovations

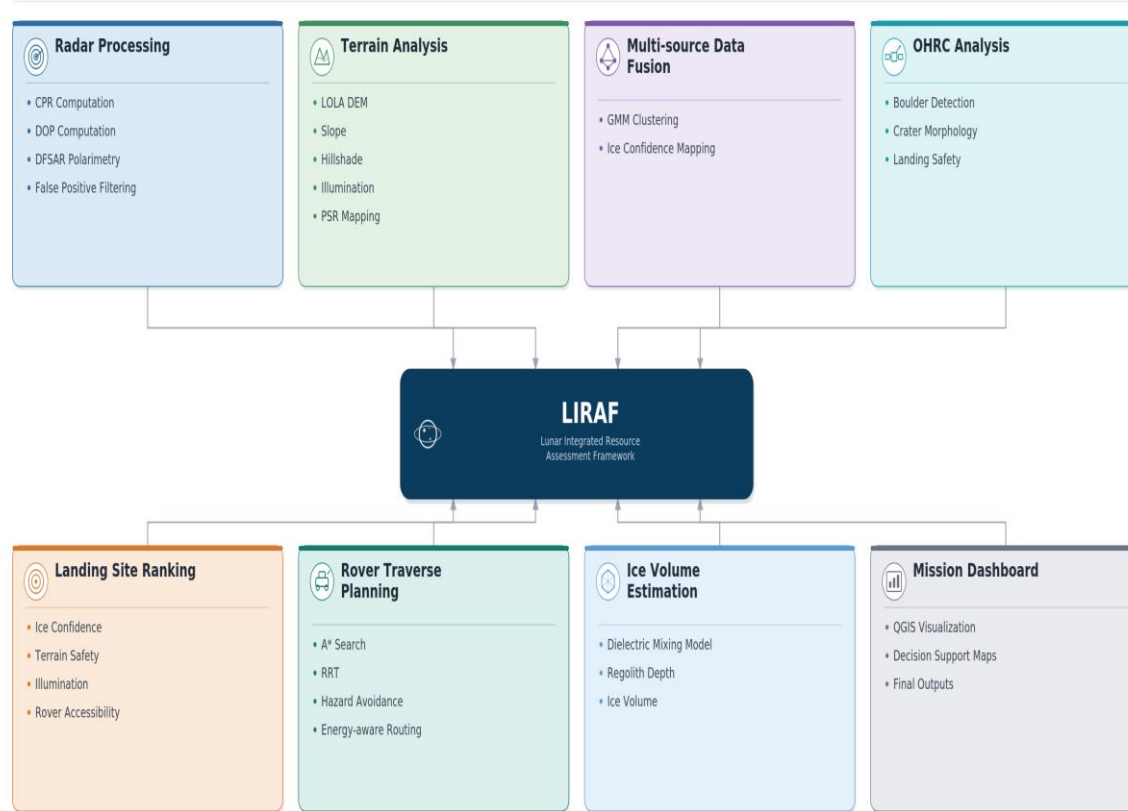
- Multi-source data fusion
- Landing site selection
- Rover traverse planning
- Scenario-based ice resource estimation
- Pre-validated Mini-RF pipeline adapted to DFSAR during finale

Features Offered by LIRAF

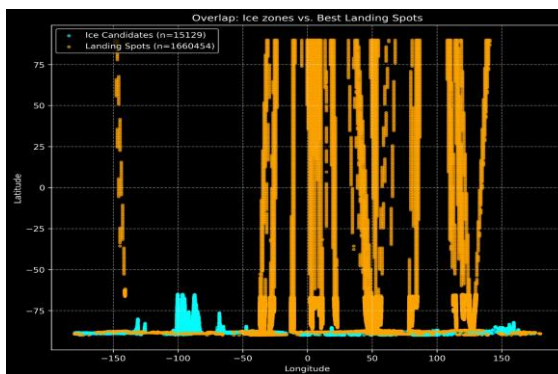
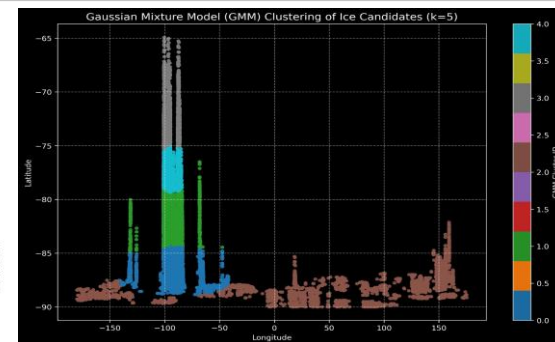
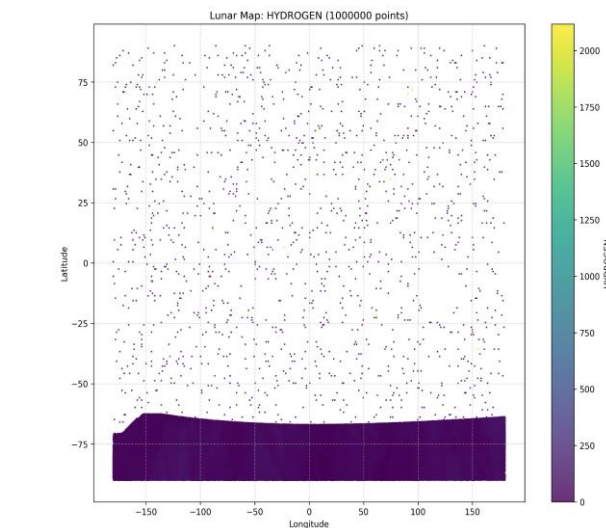
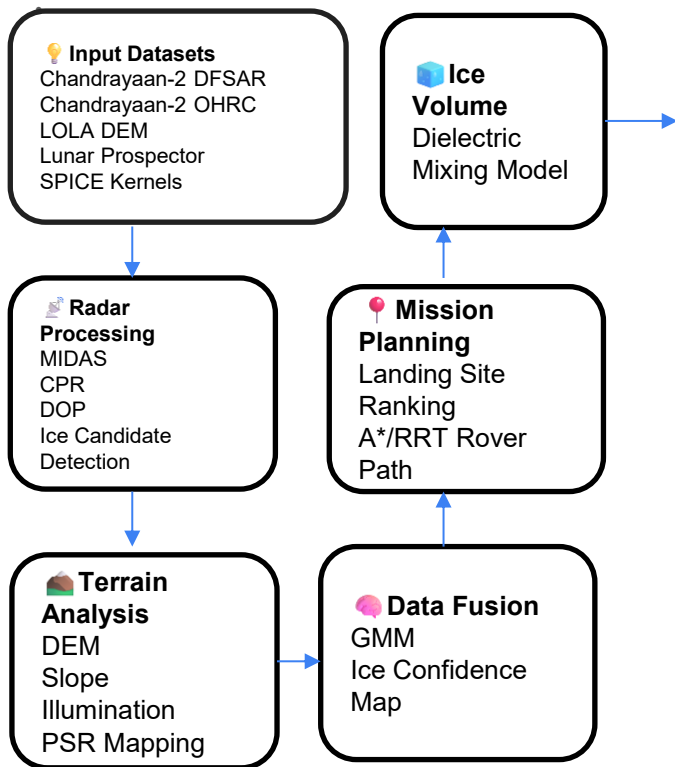
- CPR & DOP computation
- Terrain and PSR mapping
- OHRC-based landing safety assessment
- Multi-source GMM fusion
- Ice confidence map
- Landing site ranking
- A*/RRT rover path planning
- Ice volume estimation
- QGIS mission dashboard

Core Capabilities of LIRAF

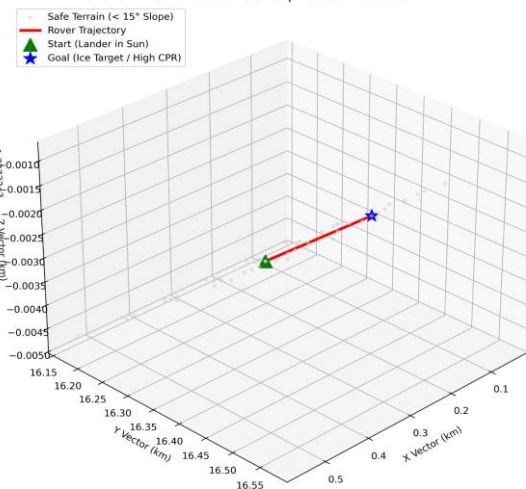
Lunar Integrated Resource Assessment Framework — Integrated Mission Architecture



LIRAF Workflow and Preliminary Results

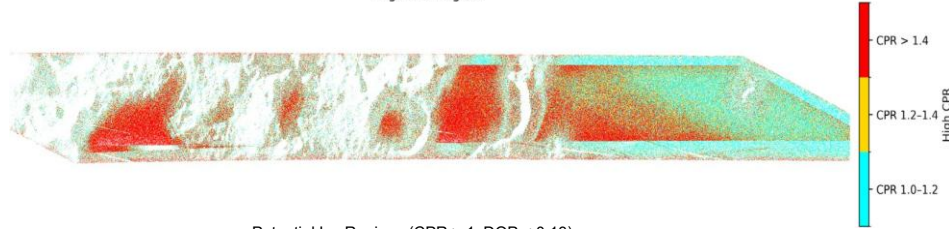


Winning Lunar Route Analysis
Kinematic Cost: 55.00 | Nodes Traversed: 2

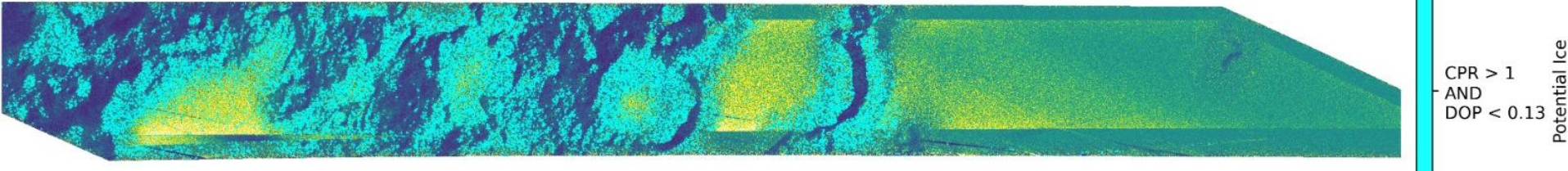


Preliminary Results Using Public Planetary Datasets

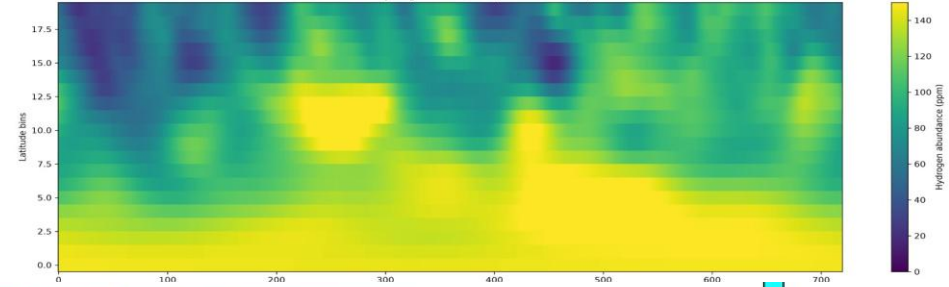
High CPR Regions



Potential Ice Regions (CPR > 1, DOP < 0.13)

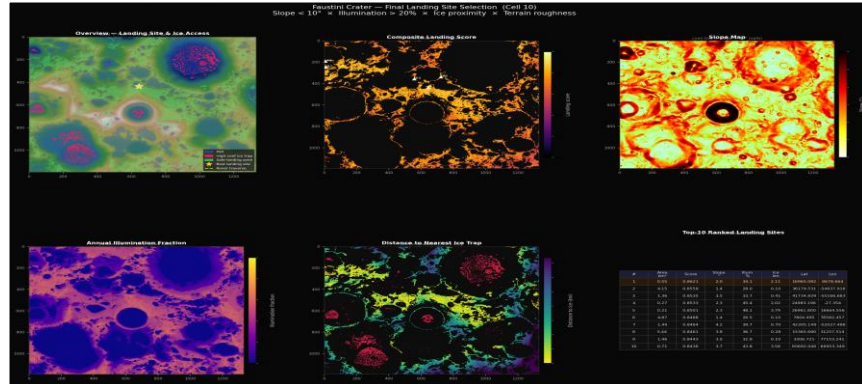


South Polar Hydrogen Abundance (Latitude < -80°)



Fastini Crater - Final Landing Site Selection (Cell 10)

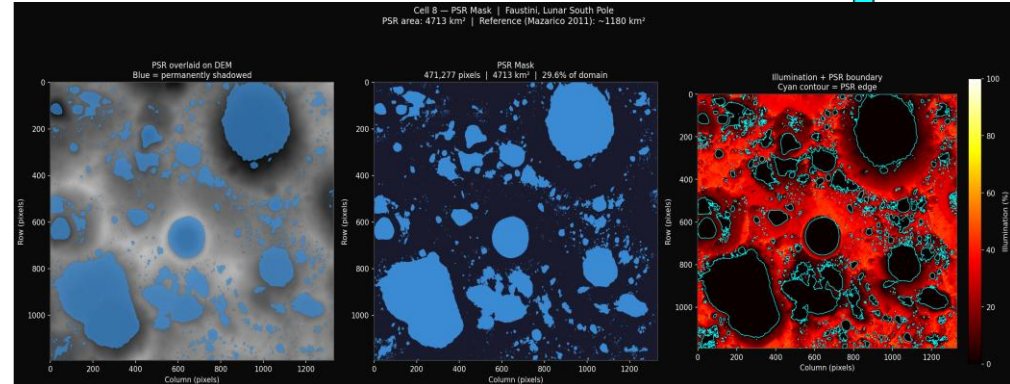
Slope = 10° ± Illumination = 20% ± Ice proximity ± Terrain roughness



Landing Site Selection Workflow

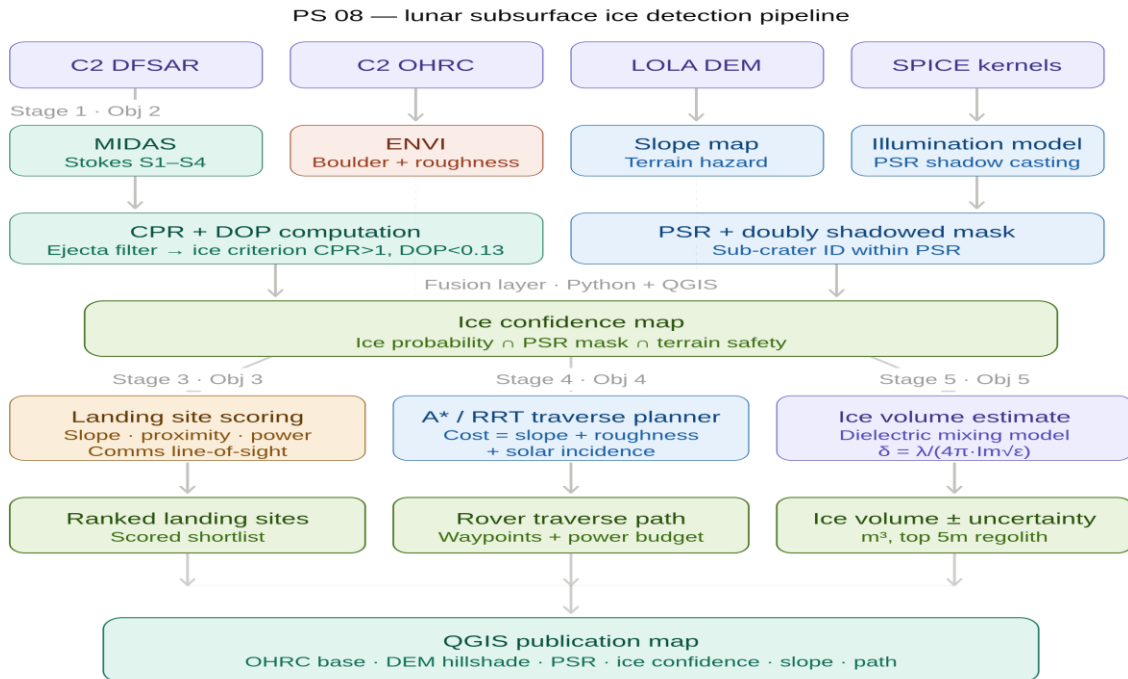
Cell 8 - PSR Mask | Fastini, Lunar South Pole

PSR area: 4713 km² | Reference (Mazarico 2011): ~1180 km²



PSR & Illumination Analysis

Architecture diagram of the proposed solution



Config-driven pipeline — dataset substitution at finale requires no code changes
 All modules pre-validated on LRO Mini-RF proxy data, Faustini crater, during 7-week prep

Technologies Used :

Category	Technologies
Programming	Python
Remote Sensing & GIS	MIDAS, QGIS, GDAL, Rasterio
Scientific Computing	NumPy, Pandas, Matplotlib
AI / Machine Learning	Scikit-learn, Gaussian Mixture Model (GMM), cuML
GPU Computing	CUDA, CuPy
Planetary Datasets	Chandrayaan-2 DFSAR, Chandrayaan-2 OHRC, LRO Mini-RF, LOLA DEM, Lunar Prospector, SPICE Kernels

Estimated Implementation Cost

Component	Estimated Cost (INR)	Remarks
Software & Libraries	₹0	Python, QGIS, GDAL, Rasterio, NumPy, Scikit-learn (Open Source)
Planetary Datasets	₹0	Chandrayaan-2 DFSAR & OHRC (Hackathon), LOLA DEM, Mini-RF, SPICE
Computing	₹0	Personal Laptop / Google Colab
Cloud / GPU	₹0	Optional Google Colab GPU
Deployment	₹0	Offline QGIS Dashboard
Total Estimated Cost	₹0	Uses open-source tools and publicly available datasets

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THANK YOU